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Characterizing the Thermal Behavior of Large Deployable Mesh Reflectors Under Spatially Nonuniform Radiation

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ABSTRACT

Large deployable mesh reflectors (DMRs) play a vital role in spaceborne communication and observation systems because of their low mass and compact stowage capability. Nevertheless, maintaining stringent surface accuracy while mitigating pronounced thermal influences in the orbital environment remains a persistent engineering challenge. This study develops a spatially discretized thermal modeling framework that incorporates direct solar radiation, Earth albedo, a prescribed equivalent Earth-infrared heat input, section-dependent solar-exposure factors, structural self-shadowing, and reflector emission to space. The proposed method is implemented to modeling of an 817-node center-feed reflector and an 849-node offset-feed reflector. Each reflector is discretized into one to 10 spatial sections and analyzed over four idealized geostationary-orbit eclipse-season cycles. Temperature distributions are evaluated at representative spatial nodes, and the influence of spatial discretization is quantified using normalized absolute temperature differences referenced to the 10-section model. For the center-feed reflector, the section-dependent temperature field is further converted into thermal strains and thermally modified stress-free cable lengths, and the resulting shape distortion of its reflecting surface is evaluated using the direct root-mean-square surface error.

1 | Introduction

Over recent decades, large deployable mesh reflectors (DMRs) have attracted sustained scholarly attention because of their low structural mass and superior packaging efficiency for spaceborne missions [1–3]. The theoretical and computational treatment of prestressed cable-network systems, originally established nearly half a century ago, has progressed in parallel with advances in DMR technology [4]. Miura [5] pioneered the interpretation of the DMR as a tension–truss configuration, adopting a spatially integrated cable lattice to generate a parabolic reflective geometry. This tension–truss antenna architecture comprised a triangular cable framework, a radio

frequency (RF) reflective mesh, and a load-bearing substructure responsible for both mechanical support and the introduction of cable pretension. The same structural philosophy was later implemented in the AstroMesh deployable antenna system [6] (see Figure 1), characterized by two ring-reinforced geodesic truss domes deployed through a single-cable actuation mechanism. Compared with alternative mesh reflector concepts, this arrangement provides notable benefits in terms of lower structural mass, minimized stowed volume, reduced surface error, decreased cost, and shortened program timelines. Nayfeh and Hefzy [7] contributed to the field by introducing geometric modeling methods for designing large lattice reflectors, focusing on subdividing and mapping

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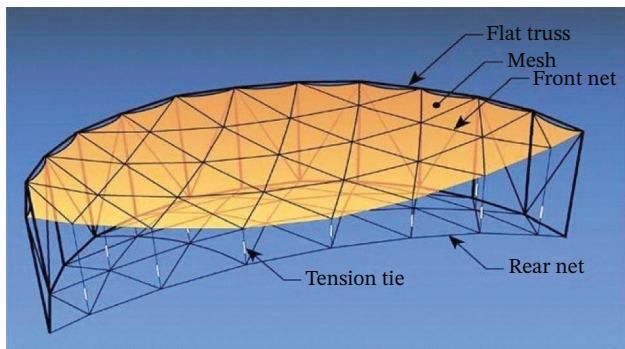


FIGURE 1 | The AstroMesh space antenna system with deployed working surface.

techniques to achieve accurate spherical and parabolic reflector surfaces. Additionally, Tibert and Pellegrino [8] proposed and demonstrated a compact deployable tensegrity reflector concept for small satellites that optimizes prestress and geometry to achieve a lightweight 3-m precision antenna stowable in a very small launch envelope.

The geometric configuration of a DMR fundamentally governs the precision of its reflecting surface as well as its overall functional performance. Consequently, constructing an accurate mesh geometry and rigorously evaluating surface accuracy constitute core tasks in DMR structural design. Shi et al. [9] introduced an optimal geometry mesh design method for large DMRs, where pseudo geodesic curves were utilized to construct the mesh configuration. This approach enhanced surface accuracy, ensured uniform member lengths, and decreased the total cumulative length of structural members. Building upon this foundation, Yuan et al. [10] advanced the methodology through the development of the projecting surface method, aimed at alleviating surface deviations arising from node placement directly on the prescribed working surface. This refinement further enhanced mesh configuration quality and contributed to improved surface accuracy. Sheth et al. [11] explored various design methods for cable mesh configurations in DMRs, presenting a force density method and an algorithm for optimal mesh generation to ensure high surface accuracy and antenna performance. Yuan et al. [12] proposed a boundary cable generation and optimization method based on a generalized topology representation combined with an optimization strategy. Their approach aimed to enhance surface accuracy and achieve uniform cable tension distributions across different design variables and reflector support configurations. Additionally, Chen et al. [13] conducted a study on the hybrid iteration of force density method for form finding of mesh reflectors in large cylindrical parabolic antennas. In that work, genetic algorithms were applied to optimize boundary cable tension, thereby improving surface accuracy and tension uniformity, while simultaneously considering elastic deformation effects inherent in supporting structures.

Throughout orbital operation, a DMR is exposed to a time-varying and spatially nonuniform radiative environment. Temporal variations in the external radiative inputs produce repeated thermal cycling, whereas spatial variations in absorbed radiative flux lead to nonuniform temperature distributions across the reflector surface. These temperature variations induce

thermal expansion and contraction within the cable members, which may result in appreciable geometric distortion. This pronounced sensitivity arises from the inherent lateral load responsiveness of cable members, primarily because they possess negligible shear and bending rigidities [14]. As a result, surface accuracy deteriorates, and the radio frequency performance of the reflector is adversely affected. To mitigate these thermally induced effects, multiple approaches have been explored in both the design phase and operational planning of DMR systems. Hedgepeth [15] was the first to investigate influence of thermal strain on surface accuracy, employing a static analytical formulation in which structural response was characterized by the average temperature change of the reflector. In that treatment, dynamic thermal contributions were assumed to remain within a factor of two relative to the corresponding static effects. Yuan and Yang [16] developed an optimal shape correction strategy for large mesh reflectors subjected to thermal distortion. Their formulation established a linear relation between nodal displacement and original member length and incorporated a mixed integer optimization algorithm to reduce the required number of actuators while preserving high surface accuracy. Guo et al. [17] performed a coupled thermal-structural finite-element investigation of a deployable reflector exposed to extreme thermal loading conditions. Their results revealed pronounced degradation in structural performance and underscored the importance of incorporating reliability and safety considerations into subsequent reflector design. Tang et al. [18] evaluated the root mean square (RMS) error of a reflector by modeling thermal strains as stochastic member length perturbations. In contrast, Yang et al. [19] represented thermal strains as a periodic phenomenon under the assumption of a spatially uniform temperature field. Xie et al. [20] conducted a theoretical investigation of dynamic shape control for DMRs based on a feedback control strategy incorporating gain scheduling, with the objective of counteracting thermally induced deformations and structural vibrations to maintain elevated surface accuracy and operational stability in the space environment. Extending the examination of on-orbit thermal influences on DMRs, Nie et al. [21] introduced an optimization-oriented design methodology for mesh reflector antennas that integrates truss deformation behavior with thermal effects, thereby reducing configuration errors arising from extreme orbital temperatures and decreasing the extent of shape correction required prior to launch. Sun et al. [22] conducted thermal analysis and developed a multimaterial optimization design for mesh reflector antennas to enhance on-orbit surface accuracy under space thermal effects. Yang et al. [23] proposed a surface shape stability design method for mesh reflector antennas, emphasizing passive precontrol of on-orbit thermal deformation by optimizing structural parameters to ensure coordinated changes in internal forces and thermal insensitivity. Tan et al. [24] previously proposed a dynamic thermal modeling approach for DMRs by incorporating solar radiation, Earth-infrared radiation, albedo radiation, and reflector self-emission into a preliminary 101-node center-feed DMR model.

Despite these advances, many reflector studies continue to represent the thermal state using a single average temperature or a prescribed static temperature change. Such assumptions cannot resolve location-dependent temperature histories or

the differential thermal strains that develop in a large cable network. The present study therefore introduces a spatially discretized transient thermal framework in which solar radiation, Earth-infrared radiation, Earth albedo radiation, and reflector emission are treated within a common energy-balance model. Section-dependent absorbed-flux factors are used to represent local differences in solar incidence, projected area, and structural self-shadowing, while all sections follow the same orbital time function. Compared with the preliminary 101-node center-feed study [24], the present work uses 817-node and 849-node reflector models, adds an offset-feed configuration, examines discretizations from 1 to 10 sections, and quantifies the sensitivity of representative nodal temperature histories. For the center-feed reflector, the section-dependent temperature field is further converted into thermal strains and thermally modified stress-free cable lengths, and the resulting equilibrium configuration of the reflecting surface is evaluated using the direct RMS error. The organization of the remaining sections is outlined as follows: The problem of thermal modeling for large DMRs, considering various heat sources and assumptions, is described in Section 2. The dominant radiation factors for thermal modeling are discussed in Section 3. The dynamic thermal modeling method, including heat conduction modeling and surface accuracy evaluation, is introduced in Section 4. Simulation results of the dynamic thermal models of a center-feed and an offset-feed large DMRs are presented in Section 5, followed by conclusions in Section 6.

2 | Problem Statement

Large DMRs operate as integral components of satellite communication systems, interacting with amplification and transmission subsystems while exposed to the external space environment [25]. A geostationary satellite orbits Earth once every 86,400s, moving at an approximate velocity of 3.1 km/s [26]. The orbit is located at an altitude of about 35,788 km and is characterized by zero inclination and zero eccentricity. Along this orbital path, the reflector experiences distinct radiative environments. During illuminated phases, solar radiation, Earth-infrared emission, and Earth-reflected solar radiation simultaneously contribute to heating. Variations in direct solar radiation alter the relative contributions of the radiative heat sources to the thermal balance of the reflector. When direct solar radiation is absent, Earth-infrared radiation remains as an external heat input, while thermal emission from the reflector provides continuous radiative cooling. Throughout the orbital cycle, the reflector continuously emits infrared radiation to deep space [27].

To establish a tractable yet physically meaningful model, the following assumptions are adopted:

1. The equivalent reflector surface is diffuse-gray. Convection is absent in vacuum, and radiation from satellite components not included in the model is neglected.
2. The radiative energy balance includes direct solar heating, Earth-infrared heating, Earth albedo heating, and thermal emission from the reflector to space.

3. Heat transfer mechanisms considered in the dynamic model consist of radiative exchange and internal conduction.
4. Owing to the shallow curvature of the reflector, the parabolic geometry is approximated as an equivalent planar disk for thermal computation.
5. Assembly and fabrication are assumed to occur at room temperature (21°C), which serves as the reference state for thermal strain calculations.

3 | Analysis of Dominant Radiation Heat Flux

In the study of the dynamic thermal response of large DMRs, four critical sources of radiation flux are identified: solar radiation, Earth-infrared radiation, Earth albedo radiation, and thermal emissions from the reflector. Solar radiation [28], being the primary source, impacts the reflector surface directly, whereas Earth-infrared radiation [29] provides an additional longwave radiative heat input. Earth albedo radiation [30], which is the reflection of solar energy from Earth, further complicates the thermal environment. Finally, the thermal emissions from the reflector play a crucial role in the overall energy balance [31]. A comprehensive understanding of these fluxes is essential for accurately predicting the thermal behavior of the reflector, ensuring performance of the reflector in harsh space environment. This section delves into the characteristics of these radiation fluxes, providing a foundation for thermal management strategies of DMRs.

3.1 | Solar Radiation Flux

Solar radiation provides the primary external heat input affecting the thermal response of a large DMR. In the present analysis, the total solar irradiance (TSI) is assumed to remain constant at $S_0 = 1361 \text{ W/m}^2$. Although TSI varies with solar activity, its variation over the four orbital cycles considered in this study is small relative to the spatial variations in absorbed solar flux across the reflector. Therefore, S_0 is treated as constant in the present transient thermal analysis [32]. However, the solar radiation absorbed by different regions of the reflector is spatially nonuniform because of differences in local surface orientation and structural self-shadowing. Previous studies have shown that self-shadowing can reduce local solar heating and produce noticeable variations in the temperature distribution and thermal deformation of orbiting truss structures [33, 34].

To represent both the temporal and spatial variations in absorbed solar radiation, the reflector surface is divided into N_{spatial} sections. The absorbed solar heat flux applied to section i is expressed as

$$q''_{\text{sol},i}(t) = \alpha_s S_0 F_{\text{orb}}(t) C_i, i = 1, 2, \dots, N, \quad (1)$$

where $q''_{\text{sol},i}(t)$ is the absorbed solar heat flux in section i , α_s is the solar absorptivity of the reflector material, S_0 is the TSI, $F_{\text{orb}}(t)$ is the common time-dependent orbital modulation factor, and C_i is the section-dependent absorbed-flux correction factor.

The correction factor C_i represents the spatial variation in effective solar exposure associated with local solar incidence angle and structural self-shadowing. It may be expressed in the general form

$$C_i = F_{sh,i} \max(0, \cos \beta_i), \quad (2)$$

$$F_{sh,i} = \begin{cases} 1, & \text{fully illuminated,} \\ 0, & \text{fully shadowed,} \\ 0 < F_{sh,i} < 1, & \text{partially shadowed,} \end{cases} \quad (3)$$

where β_i is the effective solar incidence angle, and $F_{sh,i}$ is the local self-shadowing factor, ranging from zero for complete obstruction to unity for full illumination. The term $\max(0, \cos \beta_i)$ ensures that only radiation incident on the exposed side of the surface contributes to the absorbed solar heat flux.

For a section with surface area A_i , the corresponding absorbed solar power is

$$Q_{sol,i}(t) = A_i q''_{sol,i}(t) = \alpha_s S_0 F_{orb}(t) C_i A_i. \quad (4)$$

All sections are subjected to the same temporal function $F_{orb}(t)$. The spatially nonuniform solar loading is introduced through the section-dependent factors C_i , rather than through different temporal histories for individual sections. Increasing the number of sections provides a finer spatial distribution of the prescribed absorbed solar flux.

3.2 | Earth-Infrared Radiation

Earth-infrared radiation represents to the longwave thermal radiation emitted by the Earth surface and atmosphere into space [35]. In general, the Earth-infrared heat flux absorbed by section i depends on the terrestrial infrared source intensity, the infrared absorptivity of the exposed reflector surface, and the Earth-to-section view factor. It may be expressed as

$$q''_{IR,i} = \epsilon_i \cdot F_{E,i} \cdot q''_{IR,E}, \quad (5)$$

where ϵ_i is the infrared absorptivity, $F_{E,i}$ is the Earth-to-section view factor, and $q''_{IR,E}$ is the terrestrial source intensity. In the present reduced-order APDL model, $F_{E,i}$ and ϵ_i are not evaluated separately; instead, the Earth-infrared contribution is represented by a prescribed equivalent incident flux of 341 W/m^2 . An effective infrared absorptivity of $\epsilon_i = 0.6$ is used for the reflector. This common heat input is applied uniformly to all sections and all discretization cases. During periods without direct solar and albedo inputs, the prescribed external heat flux reduces to this constant Earth-infrared contribution. Therefore, Earth-infrared heating does not generate section-to-section temperature differences in the present model.

3.3 | Albedo Radiation Flux of Earth

Earth albedo radiation represents the portion of incident solar radiation reflected by the surface and atmosphere of Earth

toward the spacecraft. Its magnitude depends on surface reflectivity, cloud coverage, atmospheric conditions, orbital altitude, reflector orientation, the illuminated portion of Earth, and Earth-to-reflector viewing geometry. In general, the albedo heat flux absorbed by section i may be expressed as

$$Q_{Albedo} = \alpha_{s,i} \cdot F_{alb,i}(t) \cdot \gamma^a \cdot S_0, \quad (6)$$

where $\alpha_{s,i}$ is the solar absorptivity of section i , $F_{alb,i}(t)$ is the time-dependent Earth albedo geometric factor, γ^a is the effective Earth albedo coefficient, and S_0 is the TSI.

In the present reduced-order APDL model, these quantities are not evaluated separately. Instead, their combined effect is represented by a prescribed equivalent absorbed albedo heat-flux history. The same albedo heat-flux history is applied uniformly to all sections and consistently across all spatial-discretization cases. During eclipse intervals, the albedo contribution is set to zero together with the direct solar input. Therefore, Earth albedo radiation does not generate section-to-section temperature differences in the present model; the spatial thermal variations are introduced through the section-dependent absorbed-solar-flux factors.

3.4 | Thermal Emissions from the Reflector

Radiation emissions describe the infrared energy released from the surface of a DMR as it undergoes thermal exchange with the surrounding space environment. The emission rate depends on surface temperature and emissivity, following the Stefan-Boltzmann law [36]. This radiation is the primary cooling mechanism for DMRs. Its magnitude is influenced by material properties, geometry, and orbital environment. Accurate emissivity modeling is critical for predicting temperature behavior and ensuring thermal stability [37]. Emissions must be evaluated alongside solar, Earth, and albedo radiation to capture the full thermal balance. By dividing the reflector into isothermal elements, radiation emission $Q_{i,DMR}$ of a DMR is given as

$$Q_{i,DMR} = \epsilon_i \sigma s_i T_i^4(t). \quad (7)$$

In this expression, ϵ_i denotes the emissivity associated with the i_{th} isothermal element of the reflecting surface. The parameter $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ represents the Stefan-Boltzmann constant. The term s_i corresponds to the radiative surface area of the element, and $T_i^4(t)$ is the absolute temperature of the section at time t .

4 | Structural Design and Dynamic Thermal Modeling and Analysis

This section presents a newly developed approach for dynamic thermal modeling of DMRs. This modeling is closely integrated with the structural design process to evaluate reflector performance in a challenging space environment. The design of a DMR involves a multistep process that requires careful integration of geometric accuracy and thermal stability during on-orbit missions (Figure 2).

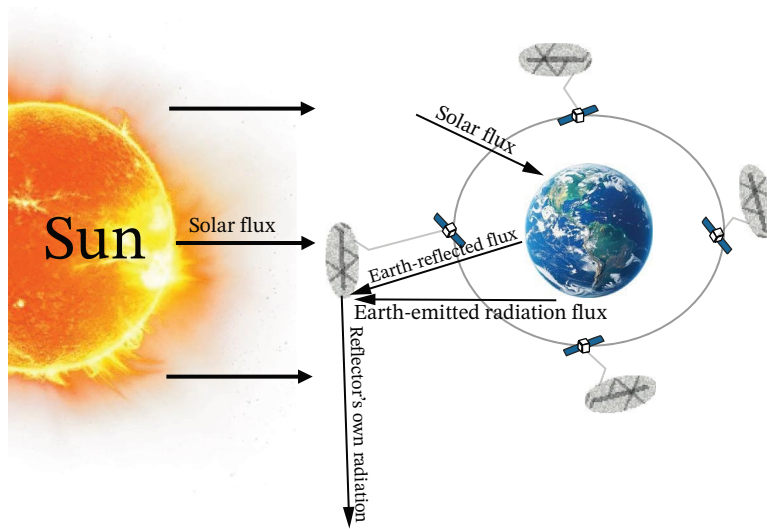


FIGURE 2 | Major thermal sources contributing to temperature variation of a DMR during the on-orbit mission.

Once the principal radiative heat inputs are incorporated into the analysis, the thermal behavior of the DMR reflecting surface is governed through a conduction-based formulation. The resulting temperature distributions in the cable members are then transformed into corresponding thermal strain fields. These thermally modified member lengths are then introduced into a dynamic relaxation scheme to determine the corresponding equilibrium configuration of the reflector after deformation. Surface accuracy of the thermally distorted geometry is quantified through the direct RMS error. By linking radiation modeling, heat transfer analysis, structural equilibrium computation, and geometric error evaluation within a unified workflow, the approach provides a practical foundation for early-stage performance assessment as well as active surface adjustment of large DMR systems operating in space environments.

4.1 | Automatic Generation of Surface Mesh Geometry and Force Finding

Prior to the dynamic thermal modeling, accurate generation of the reflector mesh geometry is required [9]. The geometric design of a DMR forms the foundation upon which its mechanical, thermal, and electromagnetic performance depends. Only after a DMR is properly designed can reliable dynamic thermal modeling and subsequent surface accuracy analysis be performed. The principal steps involved in the geometric and structural design of a DMR are described as follows:

1. For a center-feed configuration with a spherical desired working surface, the reference surface coincides with the desired working surface. In contrast, for a general center-feed parabolic DMR, the reference surface is defined as a sphere having the same aperture diameter and sag height as the target parabolic working surface. As illustrated in Figure 3a, a side view in the xz -plane shows both the desired working surface and the corresponding spherical reference surface. In this figure, Points O and A denote the center and vertex of the reference sphere, respectively, and R_s represents its radius.

For an offset-feed parabolic DMR, the desired working surface is obtained by vertically intersecting a parent paraboloid of diameter D_p and focal length F_p with a cylindrical surface, as depicted in Figure 3c. The parent paraboloid is defined in the global coordinate system (x_g, y_g, z_g) , whereas the resulting working surface, which retains the same focal length $F = F_p$, is described in the local coordinate system (x, y, z) . The origin of the local system is located at Point O . The x -axis extends from Point O toward Point C , and the z -axis is perpendicular to the plane of the elliptical boundary generated by truncation of the parent paraboloid. The focal ratio of the offset reflector is defined as $F/(2R_c)$, where R_c denotes the aperture radius. For the offset configuration, the reference surface is again spherical, as shown in Figure 3b. In this case, H_p represents the height of the working surface at the central point ($x = y = 0$). It is noted that H_p does not necessarily correspond to the maximum elevation of the working surface.

2. Mesh construction on the spherical reference surface follows a systematic procedure, as illustrated in Figure 4. The process begins by dividing the reference sphere into an even number of uniformly spaced segments using geodesic curves. This subdivision preserves axial symmetry and provides a structured framework for node placement along the geodesic lines. Nodes are distributed at consistent intervals along these curves to achieve uniform spacing over the curved surface. Next, the nodes are interconnected using curves that correspond to segments of great circles, producing a network of intersection points defined by the crossings of subdivision lines and circumferential ring lines. Additional internal nodes are introduced within each segment to increase mesh density and improve geometric fidelity. Finally, straight members are established between nodes to form triangular facets, with connectivity arranged to support efficient structural performance of the antenna system.
3. After mesh generation on the reference surface, nodes are transferred to the desired working surface. The projection strategy depends on whether the configuration is center-feed or offset-feed.

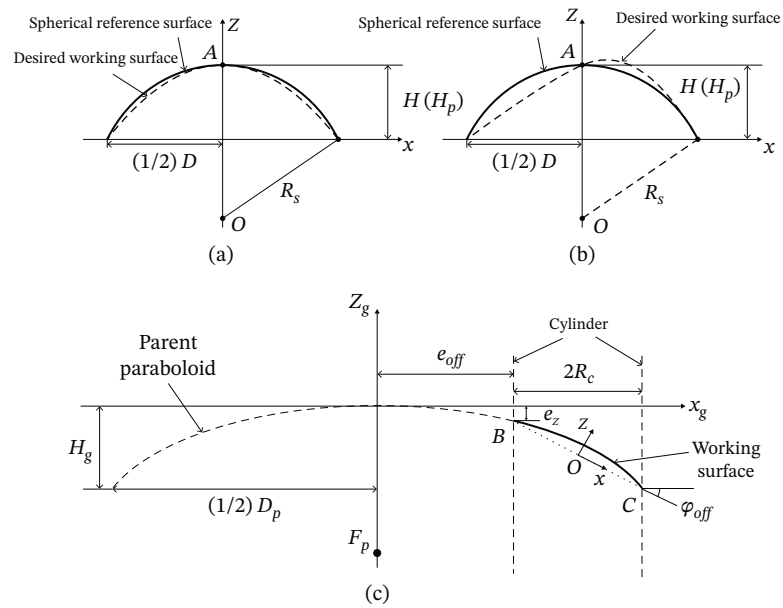


FIGURE 3 | Spherical reference surface of (a) a center-feed DMR, (b) an offset-feed DMR, and (c) the working surface of an offset-feed DMR.

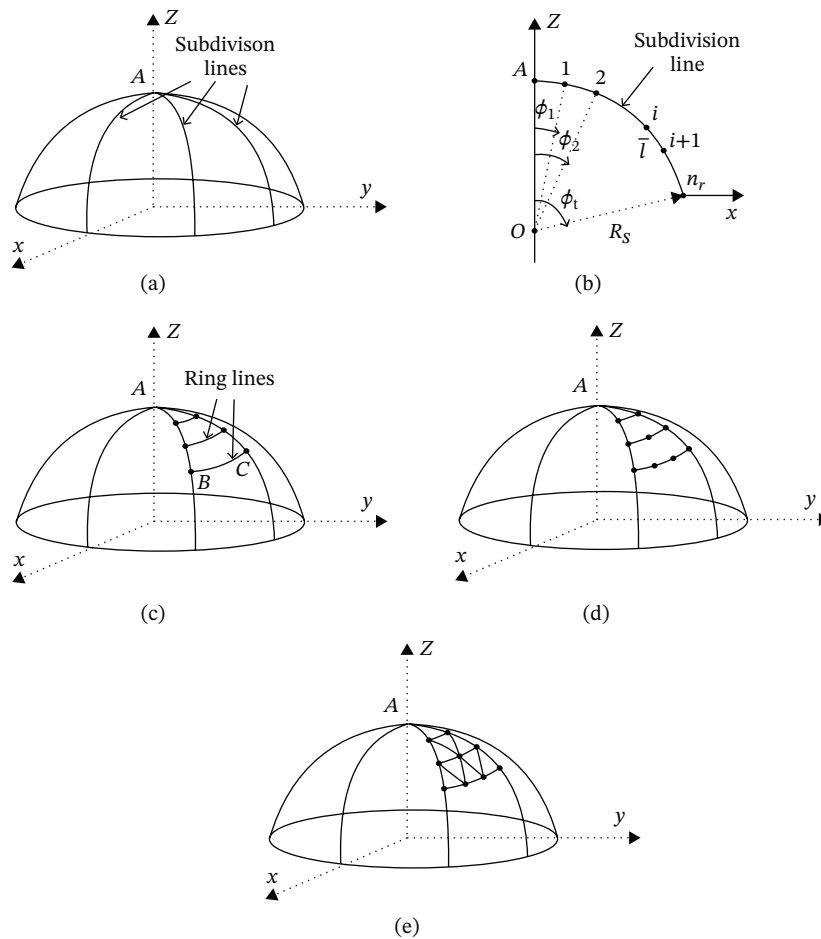


FIGURE 4 | Generation of geodesic mesh geometry on the spherical reference surface.

For a center-feed parabolic working surface, node projection is performed directly along the z -axis. During this process, x and y coordinates remain unchanged, and only the z -coordinate is

adjusted to match the target surface, as shown in Figure 5a. For an offset-feed parabolic working surface, direct projection along the z -axis is not feasible due to the elliptical aperture,

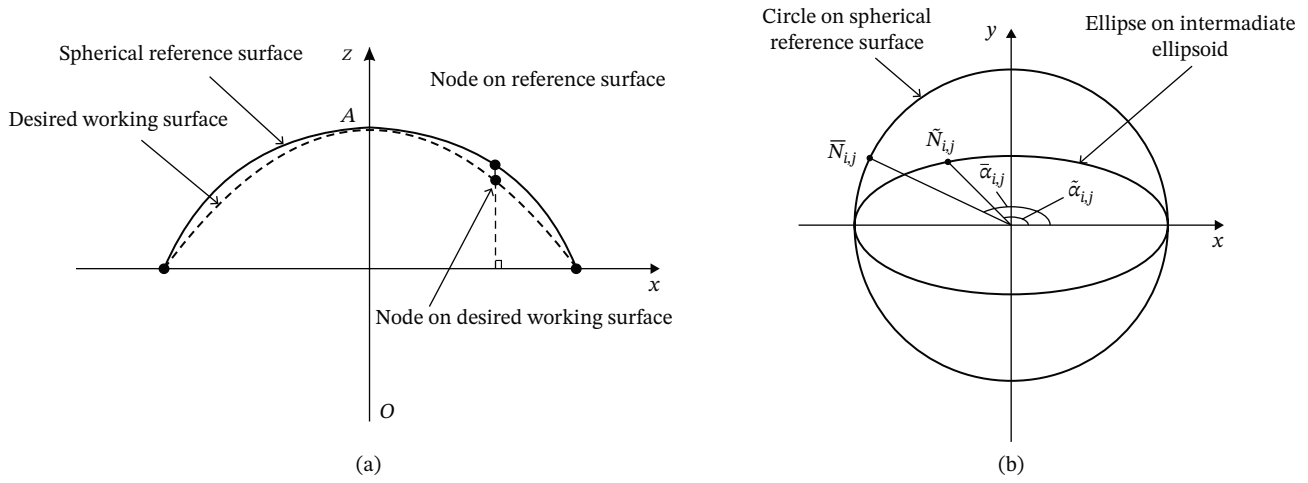


FIGURE 5 | (a) Projection of nodes from reference surface to working surface and (b) mapping from spherical reference surface to intermediate ellipsoidal surface in the xy -plane view.

which differs from the circular boundary of the spherical reference surface. Therefore, a two-stage mapping procedure is adopted. First, nodes on the spherical reference surface are mapped onto an intermediate ellipsoidal surface whose boundary conforms to the elliptical aperture of the working surface, as illustrated in Figure 5b. Subsequently, these nodes are projected onto the desired working surface along the z -axis.

For the structure of a DMR, a matrix equilibrium equation is $MT + p = 0$, where T is a vector of cable member tensions, p is a vector of external forces applied to the nodes, and M is the equilibrium matrix containing the direction cosines. In general, high tensions are desired to prevent cable slackening and improve the stiffness of the reflector structure. However, to prevent damage to cable members and preserve sufficient length-adjustment margins for active shape adjustment, the cable tension must not exceed specified operational thresholds. Therefore, assigned tensions must adhere to specific tension constraints to meet the designated design requirements. This leads to the following force constraint (Equation 8) of mixed equality and inequality:

$$MT + p = 0, \quad T_{\min} < T < T_{\max}. \quad (8)$$

Addressing the challenge of achieving an optimal tension distribution involves the use of numerical optimization due to the potentially infinite solutions of member tensions T that satisfy the force equilibrium equation in constraint (Equation 8). Given a desired tension distribution T_{des} (in DMR design, T_{des} [N] is typically specified as to be 50 L [38, 39], where L is the length [m] of the cable member), the optimization aims to minimize the discrepancy between T_{des} and the actual tensions T . This optimization task, detailed in Equation (9), can utilize methods like the active-set method or commercial software for obtaining a solution.

$$\begin{aligned} & \min f(T) \\ & \text{s. t.}, \quad MT + p = 0, \quad T_{\min} < T < T_{\max} \end{aligned} \quad (9)$$

In such an optimization problem, the objective function has many options, such as $f(T) = \|T - T_{des}\|^2$, $f(T) = \sum_{i=1}^m |T_i - T_{des,i}|$, and $f(T) = (\max(|T_i - T_{des,i}|))$, with the variables being the cable member tensions contained in vector T . The selection of objective functions and constraints is crucial, as inappropriate choices can lead to convergence issues or excessive computational demands. Recommendations for effective optimization include using a minimal number of variables, employing analytical or quadratic forms for the objective function to facilitate simpler computational processes, and minimizing nonlinearities in constraints.

4.2 | Dynamic Thermal Modeling and Surface Accuracy Analysis

During orbital service of a DMR, the temperature field within the cable members forming the reflecting surface is determined by Fourier heat conduction in conjunction with energy conservation. The transient heat transfer process is described by

$$k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) + q = \rho c \frac{\partial T}{\partial t}, \quad (10)$$

where k denotes thermal conductivity; x , y , and z represent spatial coordinates within the three-dimensional heat transfer domain; q is the radiative heat flux term; ρ is the material density of the reflecting surface; and c is the specific heat capacity. Thermal strain of the cable members ϵ_t is determined from the temperature field obtained from Equation (10) using the linear thermal expansion relation [40]:

$$\epsilon_t = \alpha_t \Delta T. \quad (11)$$

Here, α_t is the coefficient of thermal expansion and $\Delta T = T - T_0$, where T_0 corresponds to room temperature during ground assembly. The stress-free member length under thermal loading, without external mechanical forces, is expressed as

$$L_0^t = (1 + \epsilon_t) L_0. \quad (12)$$

In which L_0 is the original undeformed cable length at room temperature, and L_0^t represents the thermally adjusted undeformed length.

The dynamic relaxation method [41] serves as a computational procedure for identifying both the initial equilibrium configuration (FF-IEC) and the deformed equilibrium configuration (FF-DEC) of truss-based structures, including cable-network and tensegrity systems [4, 42]. In the present analysis, this technique is employed to compute the FF-DEC of the reflecting surface subjected to thermally induced distortion. Structural dynamic effects are neglected [43], and temperature-driven deformation is regarded as quasistatic. Each node on the reflecting surface is assigned a lumped fictitious mass, and the equilibrium configuration is obtained through dynamic relaxation with kinetic damping. The equilibrium state is obtained through numerical integration of the associated system of ordinary differential equations. The introduction of artificial damping promotes convergence of nodal motion toward a stable configuration. A refined formulation incorporating kinetic damping is adopted [44]. Within this approach, the damping coefficient is assigned a value of zero, and whenever a local peak in total kinetic energy is identified, all velocity components are immediately reset to zero to accelerate stabilization.

Once the nodal positions under thermal strain have been obtained, surface accuracy is assessed using the direct RMS error metric [45, 46]. In contrast to the nodal deviation RMS error [47]—which ignores geometric differences between individual facet planes and the corresponding best-fit surface—and to the best-fit surface RMS error [48]—which does not reflect the actual deviation of the mesh configuration from the target working surface—the direct RMS error measures the area-based deviation of the mesh geometry with respect to the specified working surface. This measure provides a consistent and reliable indicator of surface accuracy for both shallow and deep reflectors. It also permits nodal placement slightly offset from the target surface to enhance overall accuracy [10].

Consider a typical triangular facet illustrated in Figure 6, together with the corresponding desired working surface. A local coordinate system (τ, v) is established on the plane of the i_{th} facet, with the origin selected at any node belonging to that facet. Let $\mu(\tau, v)$ represent the normal separation between a point on the facet plane and the desired working surface. The deviation quantity φ_i associated with the i_{th} triangular facet is obtained by integrating the square of this normal distance over the entire facet area:

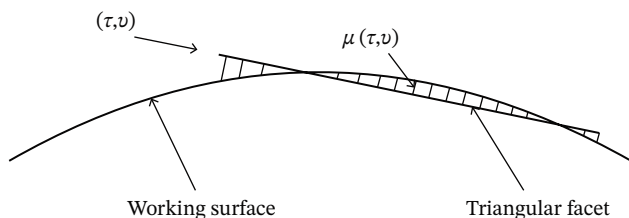


FIGURE 6 | Desired working surface and triangular facet used for evaluation of direct RMS error.

$$\varphi_i = \iint_{A_i} \mu^2 d\tau dv, \quad (13)$$

where A_i is the area of the i_{th} facet. The direct RMS error δ_{rms-d} of the entire reflecting surface is then calculated as

$$\delta_{rms-d} = \sqrt{\frac{1}{A_{mesh}} \sum_i \varphi_i} \quad (14)$$

where $A_{mesh} = \sum A_i$ representing the total mesh surface area.

5 | Simulation Results

This section presents the numerical results for the center-feed and offset-feed DMR configurations. The analysis first examines the sectional temperature distributions and their sensitivity to spatial discretization. The thermally induced surface deformation of the center-feed reflector is then evaluated using the direct RMS error.

5.1 | Thermal-Field Results and Spatial-Discretization Sensitivity

Two representative large DMR configurations are analyzed to examine the behavior of the dynamic thermal formulation. The first model consists of a center-feed parabolic reflector containing 817 nodes, whereas the second corresponds to an offset-feed configuration with 849 nodes. Both reflectors are assumed to operate in the geostationary orbital environment described in Section 3. For the center-feed case, the aperture diameter D is 17.5 m and the focal ratio (F/D) equals 1, where F denotes focal length. For the offset-feed configuration, the reflecting surface is defined by a focal length of 30 m, a focal ratio of 0.3, and an offset distance of 10 m. Within the thermal computation, the reflecting surface is idealized as a thin equivalent planar disk with a thickness of 5 mm. The cable members are fabricated from Kevlar fiber. The TSI is specified as 1361 W/m², while the effects of solar incidence are represented through the section-dependent absorbed-solar-flux factors and scattered solar radiation is excluded. Earth-infrared heating is applied as the same prescribed equivalent incident flux in every discretization case, and all cases use the same

TABLE 1 | Dimensional and material parameters of the cable members.

Parameter	Values
Young's modulus (Pa)	1.31×10^9
Linear density (g/cm ³)	1.44
Thermal expansion coefficient (K ⁻¹)	13×10^{-6}
Isotropic thermal conductivity (W m ⁻¹ K ⁻¹)	0.04
Specific heat constant pressure (J kg ⁻¹ K ⁻¹)	1380
Emissivity	0.6

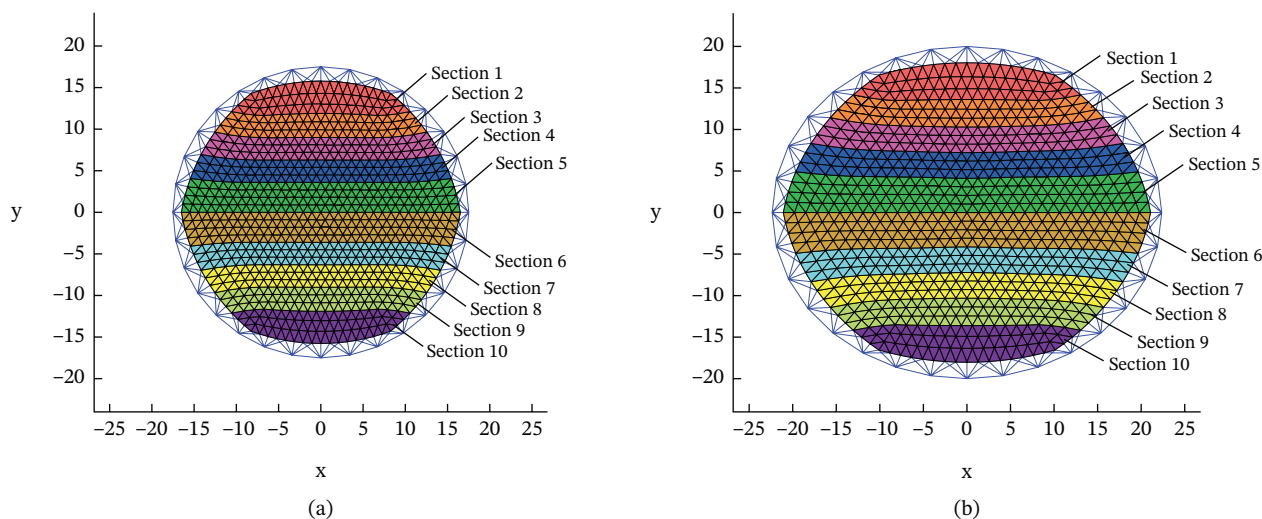


FIGURE 7 | The 10 divided sections of (a) an 817-node center-feed DMR and (b) an 849-node offset-feed DMR.

reflector-emission model. The reflector is analyzed using 1 through 10 spatial sections. For a given configuration, the orbital time function, material properties, initial temperature, time steps, and thermal boundary conditions are unchanged; only the spatial partition and section-dependent absorbed-solar-flux factors vary. As long as the reflector temperature remains above absolute zero, outward radiative emission continuously occurs until an energy balance with the surrounding space environment is achieved. Geometric and material properties of the cable members are summarized in Table 1. The proposed methodology is evaluated against the approach that neglects temperature variations across the reflecting surface. In the numerical analysis, the surface is partitioned into 10 distinct sections, as shown in Figure 7. Increasing the number of sections provides a more spatially resolved representation of the prescribed absorbed heat-flux distribution. This study conducts an analysis on sections ranging from 1 to 10. The spatially nonuniform temperature field is generated by section-dependent variations in absorbed radiative flux of the individual sections. This cyclic thermal loading generates localized temperature gradients, which subsequently influence the overall structural response of the reflector.

The spatially discretized thermal model is further examined through comparison with the uniform-temperature model. Figure 8a,b present the temperature histories of the 10 thermal sections and the corresponding uniform model for the center-feed and offset-feed DMRs over four idealized orbital cycles. All cases use the same orbital timing, material properties, initial temperature, time-step settings, and thermal boundary conditions; only the spatial partition and the section-dependent absorbed-solar-flux factors differ. The comparison illustrates the section-to-section temperature variations represented by the spatially discretized model and provides the basis for the subsequent node-level sensitivity analysis.

The simulation results derived from the thermal distribution heat maps further validate the robustness of the proposed multisection modeling framework. For the center-feed DMR, Figure 9a presents the surface temperature field at 172,740s,

where pronounced temperature gradients emerge across adjacent sections because of section-dependent variations in absorbed radiative flux. In contrast, Figure 9b shows a larger section-to-section temperature range for the offset-feed configuration at $t = 128,010$ s under the prescribed absorbed-flux factors. Table 2 compiles the temperature values across all 10 sections for both configurations, providing quantitative insight into thermal nonuniformities. Compared with the center-feed configuration, the offset-feed DMR exhibits more pronounced inter-sectional temperature differentials, leading to intensified localized thermal stresses and more intricate deformation responses. This disparity underscores the critical importance of resolving temperature fields with high spatial fidelity in both configurations. In the case of the offset-feed DMR, structural asymmetry further amplifies the nonuniformity of incident solar flux, making uniform thermal assumptions inadequate. Capturing these spatial thermal gradients with sufficient granularity is essential for accurately predicting thermally driven surface distortions and the resulting surface accuracy under orbital thermal cycling. To evaluate the sensitivity of the predicted temperature response to spatial discretization, two representative nodes located near the opposite sides of the center-feed reflector were selected. The 10-section model was used as the reference, and the normalized absolute temperature difference for an N -section model was evaluated relative to the temperature history obtained at the same physical node in the 10-section model. The 10-section curve is not displayed because its normalized difference relative to itself is identically zero.

The first node, located at $(-0.5207, 14.3006, 0.7274)$, remains within the uppermost thermal region for all discretization levels and is assigned to Section 1 in the 10-section model. Although the physical location of the node remains unchanged, the spatial extent and area of the corresponding thermal region vary with the number of sections. Consequently, the section-averaged absorbed heat flux assigned to the node also changes. As shown in Figures 10 and 11, the normalized absolute temperature difference reaches approximately 45% during the strongest thermal transition. The differences are generally smaller during the more slowly varying portions of

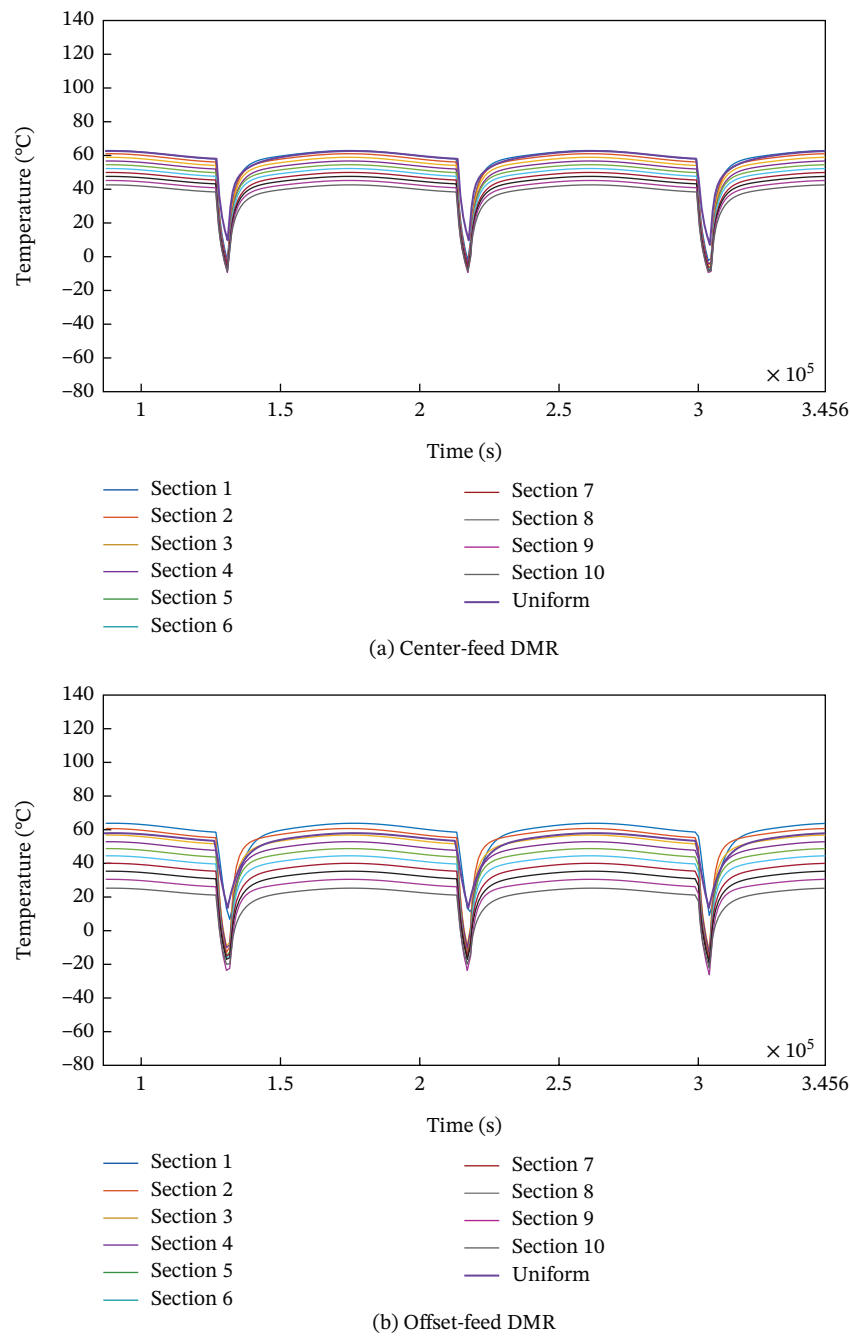


FIGURE 8 | Temperature distribution of the 10 thermal sections and the uniform model over four idealized orbital cycles: (a) the center-feed DMR and (b) the offset-feed DMR.

the orbital cycle. The difference does not decrease monotonically with increasing section number because changes in the section boundaries alter the regional heat-flux averaging associated with the selected node.

The second node, located at $(-0.5208, -14.3006, 0.7274)$, lies within the terminal region of each multisection discretization and is assigned to Section 10 in the 10-section model. The one-section model, by contrast, treats the entire reflector as a single uniform thermal region. As shown in Figures 12 and 13, the maximum normalized absolute temperature difference reaches approximately 68% for the one-section model, whereas

the peak differences for most multisection models remain below approximately 40%. These results indicate that the influence of spatial discretization on the predicted transient temperature response varies with node location.

The two comparisons characterize discretization sensitivity at fixed spatial locations. Spatial temperature nonuniformity is separately demonstrated by comparing the temperature histories of the two nodes within the same 10-section model. Because the two nodes belong to different thermal regions and experience different section-dependent absorbed heat fluxes, their temperature histories are not identical. Therefore, both

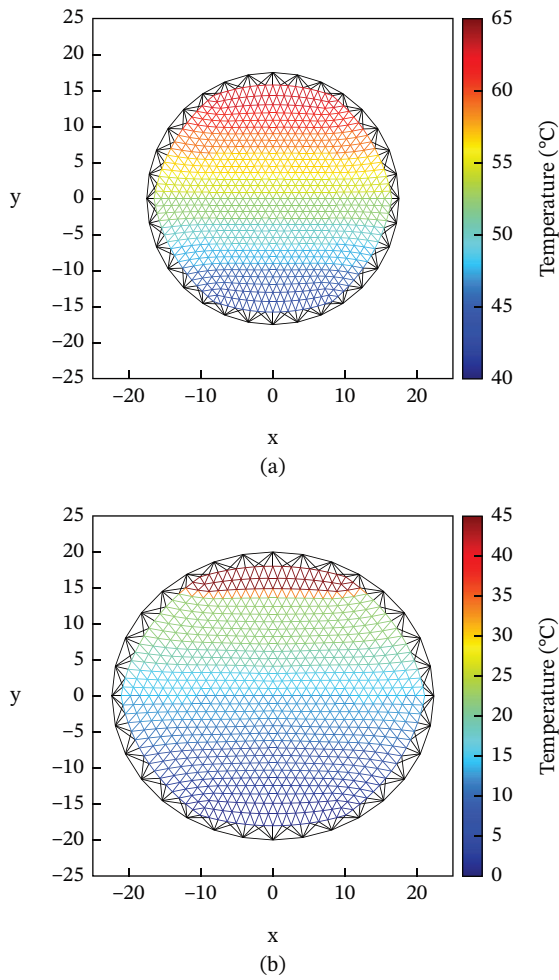


FIGURE 9 | Temperature variation heat map of (a) the center-feed DMR at 172,740s and (b) the offset-feed DMR at 128,010s.

TABLE 2 | Section temperatures of the center-feed DMR at time 172,740s and the offset-feed DMR at time 128,010s.

Thermal region	Center-feed DMR	Offset-feed DMR
	Temperature (°C)	Temperature (°C)
Section 1	62.898	44.769
Section 2	61.027	22.278
Section 3	58.899	21.798
Section 4	56.705	19.423
Section 5	54.504	15.418
Section 6	52.241	12.641
Section 7	49.915	10.889
Section 8	47.576	7.616
Section 9	45.159	2.437
Section 10	42.542	1.227
Uniform	62.583	37.423

TABLE 3 | Sectional and global direct RMS errors of the center-feed DMR at $t = 172,740$ s.

Region	Direct RMS error (mm)	RMS type
Section 1	1.55	Sectional RMS
Section 2	1.46	Sectional RMS
Section 3	1.37	Sectional RMS
Section 4	1.29	Sectional RMS
Section 5	1.23	Sectional RMS
Section 6	1.22	Sectional RMS
Section 7	1.27	Sectional RMS
Section 8	1.36	Sectional RMS
Section 9	1.48	Sectional RMS
Section 10	1.62	Sectional RMS
10-section model	1.39	Full-surface global direct RMS
Uniform model	1.20	Global RMS

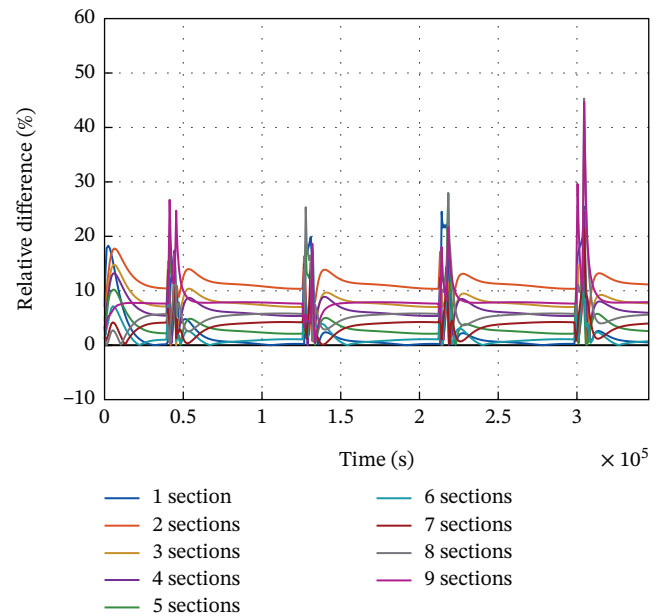


FIGURE 10 | Relative temperature difference of the center-feed DMR discretized into 1–10 sections at node $(-0.5207, 14.3006, 0.7274)$.

the spatial location of the node and the selected partitioning scheme affect the predicted transient thermal response.

Overall, the comparison indicates that the one-section model represents the uniform-temperature assumption, whereas the multisection models resolve progressively smaller thermal regions. Increasing the number of sections does not guarantee monotonic convergence at an individual node because the section boundaries and regional averaging domains change with each

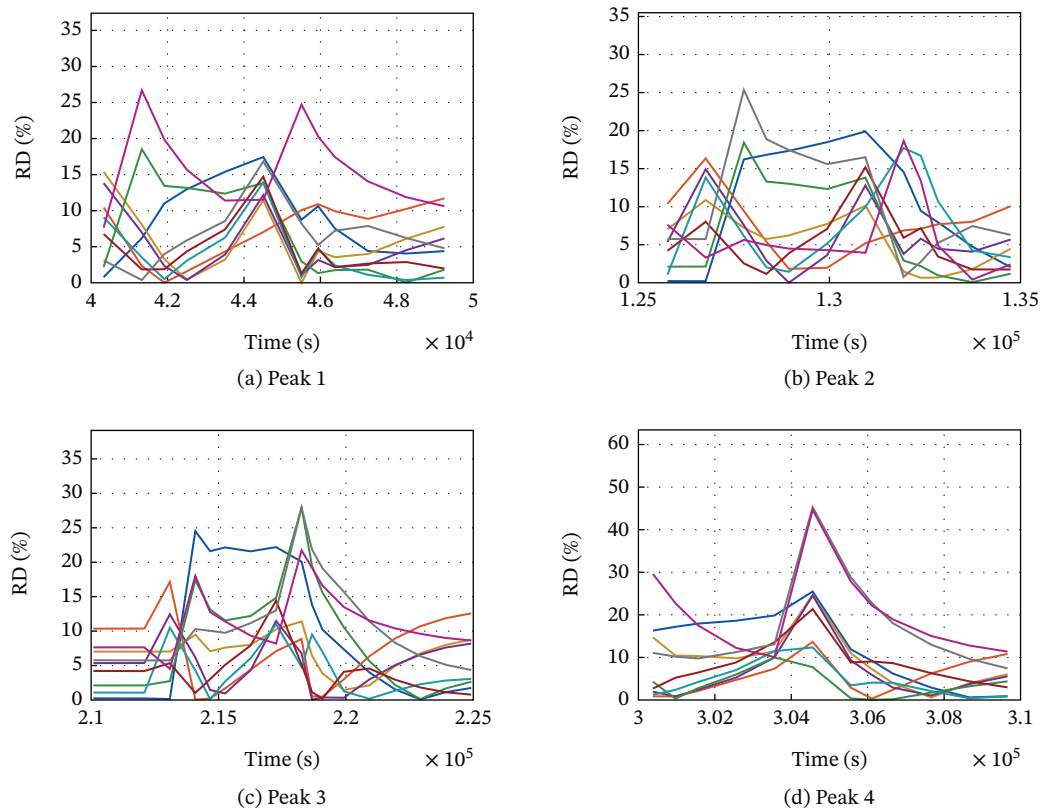


FIGURE 11 | Relative temperature difference of the center-feed DMR discretized into 1–10 sections at node $(-0.5207, 14.3006, 0.7274)$: (a) $t = 4.0 - 5.0 \times 10^4$ s; (b) $t = 1.25 - 1.35 \times 10^5$ s; (c) $t = 2.10 - 2.25 \times 10^5$ s; and (d) $t = 3.00 - 3.10 \times 10^5$ s.

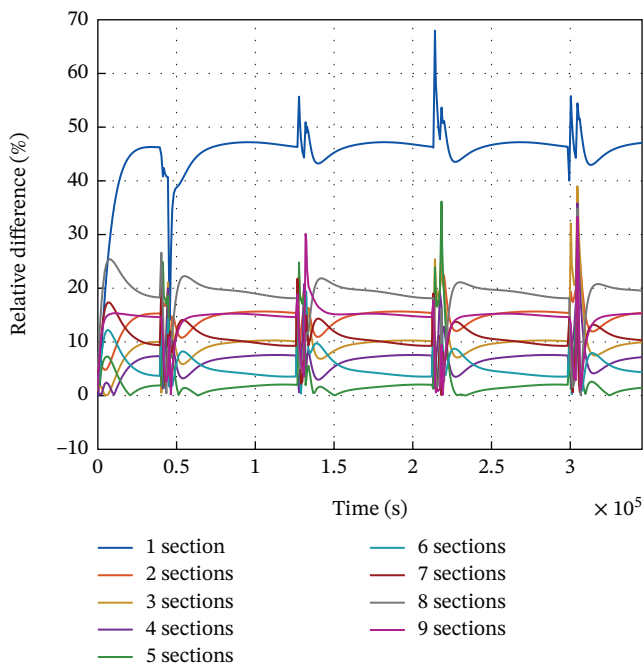


FIGURE 12 | Relative temperature difference of the center-feed DMR discretized into 1–10 sections at node $(-0.5208, -14.3006, 0.7274)$.

discretization. Nevertheless, spatially resolved models provide additional information concerning local temperature variations that cannot be represented by the uniform-temperature model.

5.2 | Direct RMS Evaluation of Thermally Induced Surface Deformation

At $t = 172,740$ s, the thermally modified stress-free cable lengths were introduced into the dynamic relaxation model to determine the equilibrium configuration of the center-feed DMR. The direct RMS error was then evaluated through area-weighted integration of the squared normal deviations between the thermally deformed triangular facets and the prescribed working surface. The sectional RMS errors ranged from 1.22 to 1.62 mm, indicating a spatially nonuniform distribution of surface error. The smallest sectional RMS error occurred in Section 6, whereas the largest value occurred in Section 10. The 10 sectional RMS values were combined using equal-section RMS weighting, resulting in an overall RMS value of 1.39 mm for the 10-section model. In comparison, the uniform-temperature model produced a global direct RMS error of 1.20 mm, which was 0.19 mm or 13.7% lower than the 10-section result. This difference indicates that suppressing spatial temperature variations can reduce the predicted differential thermal strains and consequently lead to a lower predicted thermally induced surface distortion.

The sectional RMS errors do not vary monotonically with the corresponding section temperatures because the final surface deviation is governed not only by the local absolute temperature but also by temperature differences between adjacent regions, cable orientations, initial pretension, boundary constraints, facet geometry, and cable-force redistribution during the equilibrium process. Consequently, a relatively warm region may exhibit a

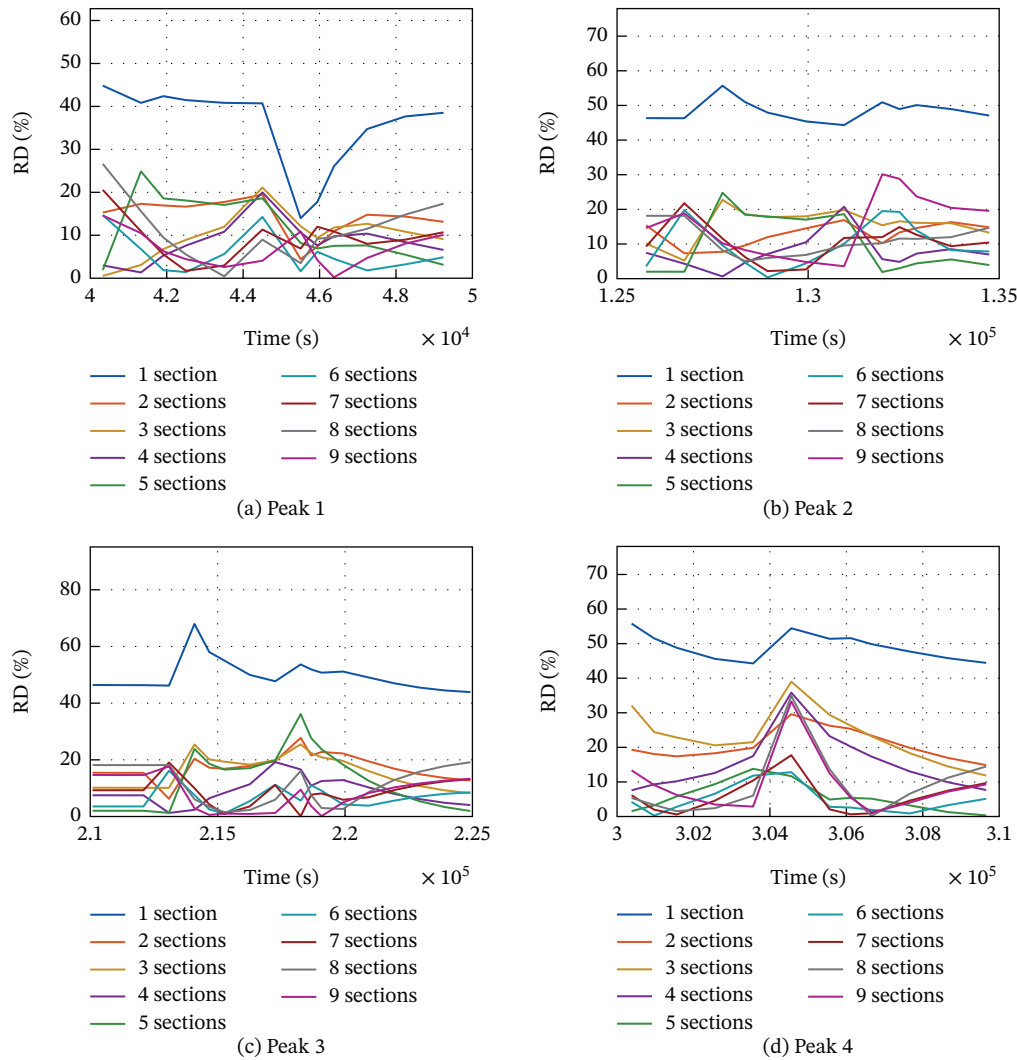


FIGURE 13 | Relative temperature difference of the center-feed DMR discretized into 1–10 sections at node $(-0.5208, -14.3006, 0.7274)$: (a) $t = 4.0 - 5.0 \times 10^4$ s; (b) $t = 1.25 - 1.35 \times 10^5$ s; (c) $t = 2.10 - 2.25 \times 10^5$ s; and (d) $t = 3.00 - 3.10 \times 10^5$ s.

smaller RMS error when its thermally induced member-length changes remain compatible with those of the surrounding cable network, whereas a cooler region may develop a larger surface deviation because of stronger differential thermal strain or local structural constraints.

6 | Conclusion

This study develops a spatially discretized thermal modeling framework for large center-feed and offset-feed DMRs subjected to multisource orbital radiation. The framework incorporates section-dependent solar-exposure factors, including local incidence effects and structural self-shadowing, to represent spatial variations in absorbed solar flux. Comparisons among the 1-section through 10-section models show that the predicted temperature histories depend on both spatial partitioning and node location. The one-section model treats the reflector as a uniform thermal region and therefore cannot represent section-to-section temperature differences. At the two selected center-feed nodes, the maximum normalized absolute temperature differences relative to the 10-section reference were approximately

45% and 68%. The 10-section model is used as a high resolution reference considered in the present study rather than as an exact converged solution. For the center-feed reflector, the section-dependent temperature field was further introduced into the dynamic relaxation model to evaluate its influence on the direct RMS surface error. These results demonstrate the importance of accounting for spatially varying radiative exposure, including structural self-shadowing, when evaluating the thermal response and surface accuracy of large DMRs.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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